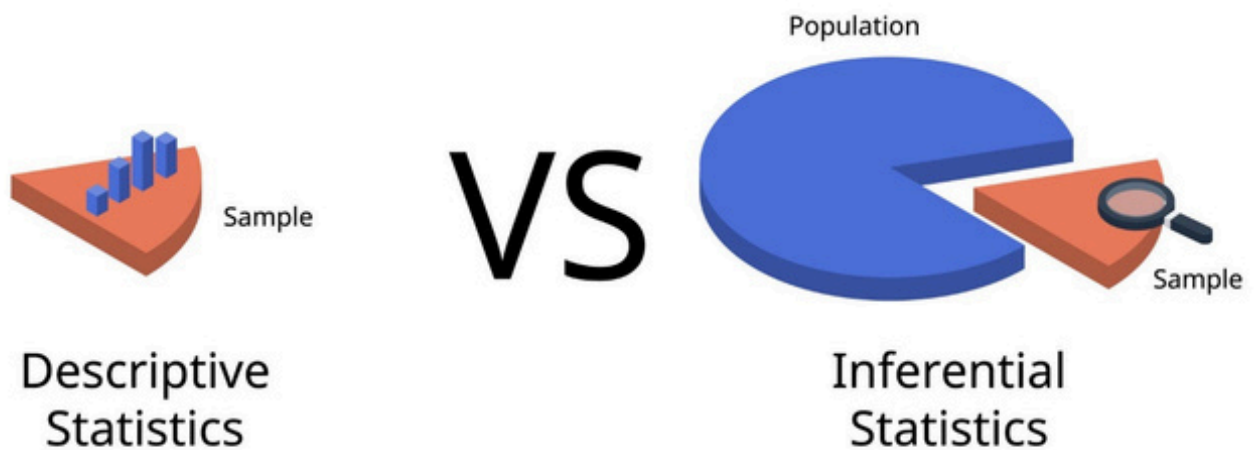


I. INTRODUCTION: THE ROLE OF UNCERTAINTY

In Machine Learning, we rarely have 100% certainty. We deal with **Stochastic** processes. Statistics allows us to quantify uncertainty, find patterns in noise, and validate whether a model's performance is statistically significant or just a result of random luck.

1.1. Descriptive vs. Inferential Statistics

- **Descriptive Statistics:** Summarizing data using Mean, Median, Mode, and Standard Deviation.
- **Inferential Statistics:** Making predictions or generalizations about a large population based on a smaller sample (e.g., predicting national election results from 1,000 surveyed voters).

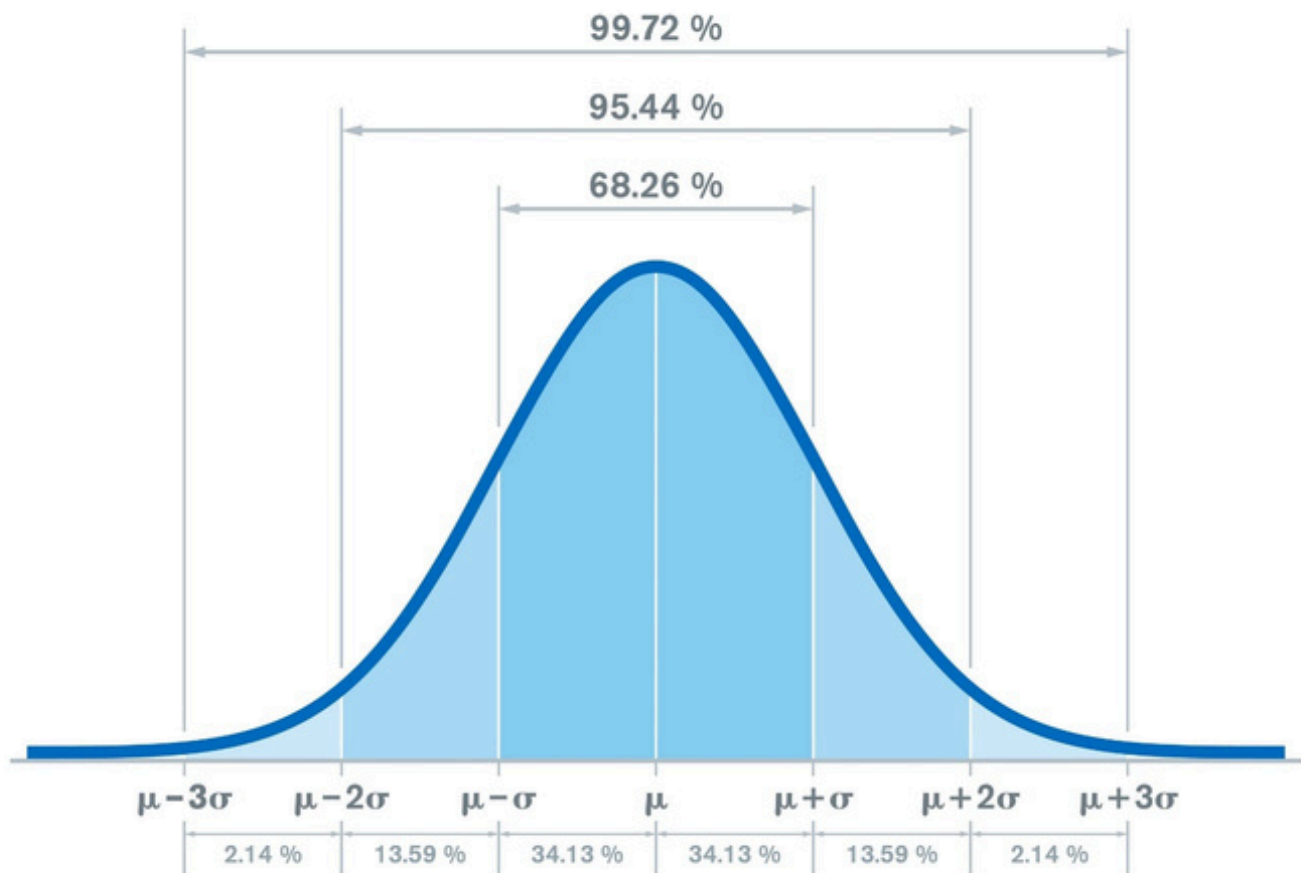


II. PROBABILITY DISTRIBUTIONS IN DATA SCIENCE

2.1. The Normal Distribution (Gaussian)

The most important distribution in nature and AI. Many Machine Learning algorithms (like Linear Regression) assume that the "residuals" or errors follow a Normal Distribution.

- **The 68-95-99.7 Rule:** 68% of data falls within 1 standard deviation (σ), 95% within 2σ , and 99.7% within 3σ .
- **Central Limit Theorem:** Explains why the sum of many independent random variables tends toward a normal distribution, regardless of the original distribution.



2.2. Bernoulli and Binomial Distributions

Used for binary outcomes (Yes/No, Spam/Not Spam, Click/No Click).

- **Application:** Logistic Regression models the probability of a "Success" (1) versus a "Failure" (0).

III. BAYESIAN STATISTICS AND CLASSIFICATION

3.1. Bayes' Theorem

Bayes' Theorem updates the probability for a hypothesis as more evidence or information becomes available.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

- **Real-world Application: Naive Bayes Classifier.** It calculates the probability that an email is "Spam" given the presence of certain words like "Free", "Winner", or "Money".

3.2. Likelihood and Prior Knowledge

- **Prior:** What we believe before seeing the data.
- **Likelihood:** How well the data supports our belief.
- **Posterior:** Our updated belief after seeing the data.

IV. HYPOTHESIS TESTING AND MODEL EVALUATION

4.1. Null Hypothesis (H_0) and P-Value

In AI Research, we use hypothesis testing to compare two models.

- **P-Value:** The probability that the observed results happened by chance. In most scientific fields, a **P-value** < 0.05 is considered "Statistically Significant".
- **Type I Error (False Positive):** Rejecting a true null hypothesis (e.g., telling a healthy person they have a disease).
- **Type II Error (False Negative):** Failing to reject a false null hypothesis (e.g., telling a sick person they are healthy).

4.2. Correlation vs. Causation

- **Pearson Correlation (r):** Measures the linear relationship between two variables (from -1 to 1).
- **Critical Note:** Just because two variables move together (Correlation) does not mean one causes the other (Causation).



V. PRACTICAL TOOLS IN THE DTU LAB

Students will use Python libraries to perform statistical analysis:

Python

```
import scipy.stats as stats

import numpy as np

# Generating normal distribution data

data = np.random.normal(loc=0, scale=1, size=1000)

# Performing a T-Test to compare two groups

t_stat, p_val = stats.ttest_ind(group_a, group_b)

if p_val < 0.05:

    print("The difference is statistically significant!")
```